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INTRO SESSION · 2026

Langfuse 101

Observability, prompt management, and evaluation for LLM applications.

01



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THE PROBLEM

LLM Observability

Incorrect output	Why did the summary change a deadline?
Slow response	Which step took the longest?
Unexpected cost	How many tokens did the request consume?
Changing quality	Did the new prompt produce a better summary?

The final response shows the result. **Observability reveals the recorded execution.**



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THE CORE IDEA

Recording application activity



Processes the meeting notes and calls the model.

Captures inputs, outputs, timings, and errors.

Organizes the recorded activity for inspection.

Framework integrations and custom instrumentation capture the steps that matter.

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THE DATA MODEL

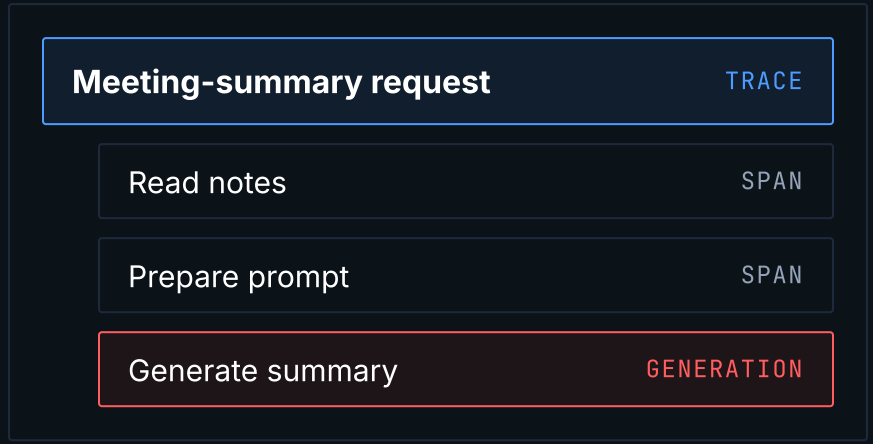
One request, multiple steps

Trace
The recorded work for one request.

Observation
An individual step within the trace.

Span
An application operation.

Generation
A model call, including its prompt and response.



Example: a summary request contains reading the notes, preparing the prompt, and generating the summary.

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DEBUGGING AND PERFORMANCE

Inputs, latency, and cost

Inputs and outputs	What each step received and produced.
Duration	Where the application spent time.
Errors	Which operation failed.
Tokens and estimated cost	How much the model call consumed.

EXAMPLE

The notes say "Friday." The summary says "Monday."

Inspecting the model input helps locate where the deadline changed.

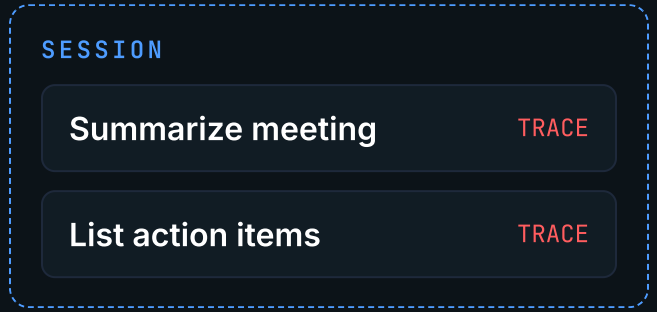
Successful execution does not guarantee a correct answer.



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ORGANIZING ACTIVITY

Related requests, shared context



Session
Groups related requests, such as a summary and its follow-up questions.

User ID
Identifies who made the request.

Tags and metadata
Describe the task, document, or application version.

Environment
Distinguishes development and production activity.

Sessions organize traces. The application manages conversational memory.

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PROMPTS AND VERSIONS

Instructions with version history

Version 1 production

"Summarize these meeting notes."

Version 2 candidate

"Summarize in three bullets. Preserve action owners and deadlines. Include only information in the notes."

Versions
Preserve changes to the instructions.

Labels
Select a version, such as production or candidate.

Trace links
Connect an answer to its prompt version.

Playground
Try prompts and models interactively.

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EVALUATION

Quality needs a criterion

Score
A judgment attached to a recorded execution.

Methods
Human review, deterministic checks, or an LLM judge.

EXAMPLE CRITERION

Does the summary preserve the deadline in the meeting notes?

Summary deadline: Monday
when the notes say Friday
score **0**

Summary deadline: Friday
matching the notes
score **1**

A score measures the criterion you define.

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REPEATABLE TESTING

Comparing versions on the same cases

Dataset
Reusable inputs and expected results.

Experiment
Runs an application version against those inputs.

Comparison
Reviews quality, latency, and cost.



The same test cases make changes easier to assess.